Nathan Wong Shih Hao 黄诗豪

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Summary

Software and AI engineer who turns LLM prototypes into secure, tested enterprise products, most recently a GenAI risk assistant for ~3,000 insurance users. Built AI and data systems at Capgemini, Hitachi Rail and TRM.

Education

The Hong Kong University of Science and Technology, Hong Kong — September 2022 to June 2026
Bachelor of Engineering in Computer Science (Exchange to University of Waterloo)
• **Awards:** Reaching Out Award (2024), University Entrance Scholarship (2022)

Work Experience

Capgemini Hong Kong | Technology Consultant | Hong Kong — March 2026 to Present
• Co-delivered a GenAI risk assistant for a large insurer's Group Risk function, built to serve ~3,000 users, from prototype to user acceptance testing. Staff ask risk questions in plain English and get narrative, tables and charts.
• Owned the entire chat frontend as sole developer, writing 15k lines of TypeScript in Next.js and React. Streamed agent answers live, with multi-thread chat history, inline charts and CSV export so users can act on results.
• Made the assistant enterprise-ready: secured access with Microsoft Entra ID, routed all traffic through a hardened proxy to the private backend, and shipped via a GitHub Actions CI/CD pipeline to Artifactory and Azure.
• Unblocked deployment by fixing firewall and routing faults with the infrastructure team. Stopped long LLM answers from failing at the App Service 230-second idle timeout by adding 15-second SSE heartbeats to the stream.
• Raised release confidence with 64 Playwright BDD scenarios, 73 unit tests and a 48-question UAT suite that the assistant answered in full. Fed each round of UAT feedback back into chatbot tuning and the text2SQL data catalogue.
• Benchmarked 5+ SLMs (Phi, Gemma, Qwen) against LLMs on latency, memory and hallucination to shape the team's edge AI strategy. Turned the findings into a FastAPI SLM/LLM router that cut inference cost by 30%.

Hitachi Rail GTS | R&D Software Intern | Hong Kong — June 2025 to February 2026
• Developed ARIMA, SARIMAX and LightGBM forecasting models for Project NAIA, averaging 97% R^2 on forecasts.
• Engineered a Kafka pipeline with a producer-consumer schema registry and validation, unifying multi-format ingestion.
• Improved the primary scripting algorithm's performance by 20% and sped up data collection with ClickHouse batching.
• Moved Project NAIA toward production: implemented Jenkins CI/CD for integrated testing and coordinated its deployment to Hitachi's ALVEA platform on a Kubernetes cluster that handled operational management and scaling.

Total Rehabilitation Management (TRM HK) | Software Developer | Hong Kong — June 2024 to August 2024
• Evaluated multiple LLMs (GPT-4o, GPT-4o mini, Claude and others) to filter and prioritize insurance claims.
• Developed an internal staff chatbot, increasing engagement from 2 messages per week to 20 messages per day (70x).
• Designed and implemented a client-side search bar with JavaScript, C# and MySQL, reducing search time by 50%.

Leadership & Activities

HKUSTSU Southeast Asian Students' Association | President | Hong Kong — February 2023 to February 2024
• Grew event turnout by 300% across 4 Southeast Asian country groups and represented members at the Students' Union.

ICMS | 13th Internal Affairs Executive Director | Hong Kong — February 2023 to February 2024
• Led 28 chapter directors and 92 associates worldwide; built relations with Axiata, Shopee, Shell, P&G and Maybank.

Projects

Morning Digest Using GitHub Actions
• Engineered an AI pipeline (GitHub Actions, Google OAuth, Gmail, Drive) for daily briefs, lifting productivity by 30%.

Containerized Local Small Language Model Inference Using Ollama
• Deployed quantized GGUF SLMs with Ollama in Docker for reproducible, efficient inference on consumer hardware.

Time Series Forecasting
• Evaluated ARIMA, SARIMA, ETS, LightGBM and LSTM models in a self-directed S&P 500 forecasting project.

Skills and Competencies

• **Languages & Frameworks:** Python, TypeScript, JavaScript, Java, C/C++, C#, Scala, SQL, React, Next.js, FastAPI.
• **AI & Data:** PyTorch, Scikit-Learn, XGBoost, LightGBM, LLM evaluation, text2SQL, Kafka, Spark, ClickHouse.
• **Cloud & DevOps:** Azure (App Service, Entra ID, Fabric), AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions.
• **Tools:** Git, Playwright, Vitest, Artifactory, Ollama, LM Studio, Hadoop, Avro, Pulsar, Tailwind, Bash, HTML/CSS.
• **Languages:** Fluent in English, Bahasa Indonesia, Chinese (Mandarin). Work rights in HK and Malaysia, no sponsorship.
• **Interests:** Golfing, badminton, travelling, landscape photography, aviation, 3D printing, Southeast Asian economies.